llmXive Automated Review of Scaling the Horizon, Not the Parameters: Reaching Trillion-Parameter Performance with a 35B Agent

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong comparative claims against "1T-parameter models" and specific baselines that require verification. The abstract and introduction assert that Agents-A1 outperforms Kimi-K2.6 and DeepSeek-V4-pro on a list of benchmarks. While Table 1 supports the claim for SEAL-0, the text generalizes this to "outperforms 1T models" without acknowledging that the model loses to GPT-5.5 (another 1T+ class model) on BrowseComp and XBench-DS. This is a minor overstatement of the results.

More critically, the evaluation section relies on baselines that appear to be hallucinated or future-dated. Specifically, "GPT-5.5" is cited as a baseline with a specific configuration ("xhigh reasoning effort"). As of the current public record, GPT-5.5 does not exist, nor does the "xhigh" parameter. Similarly, the citation `zhang2026molclaw` (2026) and the general context of "GPT-5.5" and "DeepSeek-V4-pro" suggest the paper may be projecting future models as current baselines. If these models do not exist, the reported comparisons (e.g., GPT-5.5 scoring 84.4 on BrowseComp) are fabricated or based on hypothetical data, which invalidates the core claim of "reaching trillion-parameter performance" by comparison. The authors must verify the existence of these baselines and the accuracy of the reported scores. If these are hypothetical, the paper must clearly state that these are projected or simulated results, not empirical comparisons.

Additionally, the claim that the model "outperforms" 1T models on HiPhO and FS-O is supported by Table 1 (46.4 vs 41.1/38.7 and 79.0 vs 73.0/76.0), but the comparison against GPT-5.5 (43.3 and 78.0) shows the model is competitive but not strictly superior on FS-O (79.0 vs 78.0 is a narrow margin, and 46.4 vs 43.3 is a win). The phrasing "outperforms 1T models" is slightly imprecise but acceptable if qualified. The primary issue remains the non-existence of the GPT-5.5 baseline and the "xhigh" parameter, which undermines the credibility of the entire comparative table.

Action items.

- [**writing**] The paper makes several strong comparative claims against "1T-parameter models" and specific baselines that require verification. The abstract and introduction assert that Agents-A1 outperforms Kimi-K2.6 and DeepSeek-V4-pro on a list of benchmarks. While Table 1 supports the claim for SEAL-0, the text generalizes this to "outperforms 1T models" without acknowledging that the model loses to GPT-5.5 (another 1T+ class model) on BrowseComp and XBench-DS. This is a minor overstatement of the results

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens *runs on a vision-capable model.*

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively displays benchmark performance across multiple tasks with clear numerical annotations, but the caption is a placeholder filename and the legend relies on icons without explicit textual definition in the caption.

Figure 2

The figure effectively visualizes the optimization trajectory with clear annotations, but the caption contains a grammatical error and extraneous filename text that should be cleaned up.

Figure 3

The figure is visually clear and the panels are well-organized, but the caption text is truncated, cutting off the description for panel (d). Additionally, the y-axis labeling in panel (e) is non-standard.

Figure 4

Figure 4 provides a high-level overview of the training pipeline but omits key connections between data queries and the loss function, and fails to include 'Agent Tasks' in the query section despite listing it in the data sources.

Figure 5

The figure provides a clear and detailed visual overview of the knowledge-action infrastructure, effectively mapping the flow from data sources to the KAG and self-play loops. However, the caption contains a grammatical error omitting the model name, and minor terminology inconsistencies exist in the data source labels.

Action items.

- **[writing]** Figure 1: The caption 'Benchmark performance of [Fig1_a1_benchmarks_altair_grid-8.pdf]' is a filename placeholder rather than a descriptive summary of the data shown.
- **[writing]** Figure 1: The legend at the top uses icons that are not explicitly defined in the caption, requiring the user to infer the mapping between the icons and the model names (e.g., 'Agents-A1', 'Qwen3.6-35B-A3B').
- **[writing]** Figure 2: The caption contains a grammatical error ('Optimization trajectory of on the...') and includes a raw filename ('[appendix_mle.png]') that should be removed.

- [**science**] Figure 2: The x-axis label 'Time (hours)' is ambiguous; the caption mentions 'wall-clock time' but does not specify if this is elapsed time or cumulative compute time, which is critical for interpreting the '12-hour run' claim.
- [**writing**] Figure 3: The caption text is truncated at the end ('with tria [appendix_earth.png]'), cutting off the description for panel (d).
- [**science**] Figure 3: Panel (e) y-axis labels use cardinal directions (N, S, E, W) alongside degrees (0°, 90°, etc.), which is non-standard and potentially confusing for a heading scale (0-360°).
- [**writing**] Figure 3: Panel (d) legend includes 'Speed increase' and 'Speed decrease' markers, but the caption does not describe these specific annotations.
- [**science**] Figure 4: The diagram shows a loss function formula at the bottom, but the caption does not explain what the formula represents or how it relates to the 'salient vocabulary alignment' mentioned.
- [**writing**] Figure 4: The title 'Multi-domain Data' lists 'Search', 'Science', 'Engineering', 'Agent Tasks', and 'Inst. Following', but the diagram's query boxes only show 'Search data', 'Science data', 'Engineering data', and 'Inst. data' — 'Agent Tasks' is missing from the query section.
- [**science**] Figure 4: The flow from 'Queries of ... data' boxes to the loss function lacks clear arrows or labels indicating how these queries feed into the training process described in the caption.
- [**writing**] Figure 5: The caption contains a grammatical error ('infrastructure of .') where the model name is missing.
- [**writing**] Figure 5: The 'Training corpus' section lists 'Data base' with a space, which is non-standard terminology compared to 'Database'.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-written for a technical audience, but there are a few instances where notation and acronyms are introduced without sufficient definition for a competent reader from an adjacent field (e.g., a researcher in NLP reading a systems paper, or vice versa).

Specifically, in Section 2.1, the symbol $\mathcal{B}_d$ appears in the definition of the Knowledge-Action Graph ("Given a domain $\mathcal{B}_d$") but is never explicitly defined as a set, a label, or a specific entity type. The reader is left to infer its meaning from context, which breaks the self-contained nature of the definition. Similarly, in Section 2.2, the symbol $\perp$ is used in the trajectory equation to represent a state where no verifier fires, but this convention is not explained in the prose, forcing the reader to guess the semantic meaning of the symbol.

Additionally, the acronym “SVA” (Salient Vocabulary Alignment) is used frequently in Section 2.3 but is not explicitly expanded with the acronym in parentheses at its very first mention in the main text (it appears in the section title and later in the text, but the immediate expansion is missing in the introductory sentence of the subsection). While “OPD” is clearer, ensuring all custom acronyms are expanded at first use is a standard requirement for accessibility. Finally, in Section 3.1, the hyperparameter λ_{neg} is introduced in an equation without a preceding textual definition of what it controls, which is a minor but notable omission for reproducibility and clarity.

These issues are easily fixable with one-sentence additions or parenthetical expansions and do not detract from the core scientific contribution, but they do create small friction points for the “adjacent-field PhD” reader.

Action items.

- [**writing**] Section 2.1 (Eq. 1) introduces the symbol $\mathcal{B}_d$ in the phrase ‘Given a domain $\mathcal{B}_d$‘ without defining it. The subsequent definition of $\mathcal{G}_d$ uses $\mathcal{C}_d, \mathcal{A}_d, \mathcal{O}_d, \mathcal{V}_d$ but never explicitly states what $\mathcal{B}_d$ represents (e.g., ‘the set of tasks in domain d’ or ‘the domain label’). Define $\mathcal{B}_d$ at first use.
- [**writing**] Section 2.2 (Eq. 2) introduces the symbol ot in the trajectory definition ($v_t \in \mathcal{V}_q \cup \{ot\}$) without a textual explanation. While common in logic, in this context it is unclear if it means ‘no verifier fired’, ‘verification failed’, or ‘undefined’. Add a brief clause: ‘where ot denotes the absence of a triggered verifier’.
- [**writing**] Section 2.3 introduces ‘SVA’ (Salient Vocabulary Alignment) and ‘OPD’ (On-Policy Distillation) as acronyms. While ‘OPD’ is defined in the section title, ‘SVA’ is only introduced as a phrase in the text without the acronym expansion in parentheses at first use (e.g., ‘salient vocabulary alignment (SVA)’). Ensure both are explicitly expanded at first occurrence in the body text.
- [**writing**] Section 3.1 (Eq. 10) uses the symbol λ_{neg} in the advantage equation without defining its value or role in the text preceding the equation. The text mentions ‘asymmetric design’ but does not explicitly state that λ_{neg} is a weighting coefficient for the process reward on negative samples. Define λ_{neg} where it first appears.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent argument for scaling agent horizons via a three-stage training pipeline. However, there are specific inconsistencies between the textual claims and the data presented in the tables that break the internal logical flow of the results section.

First, in Section 4.2.1 ("Results of Domain Teacher Training"), the text explicitly states: "The most notable improvement is observed on GAIA, where the score increases from 59.8 to 85.4 (+25.6)." This claim is directly contradicted by Table 5 (`tables/search_rl.tex`), which lists the Search-enhanced Teacher's GAIA score as **95.1**, not 85.4. The difference (95.1 - 59.8 = 35.3) is significantly larger than the claimed 25.6. This is a clear non-entailment where the conclusion (the specific numbers cited) does not follow from the evidence (the table provided in the same section).

Second, the text in Section 4.2.2 references "Table 6, Table 7 and Table 8" when discussing the comparison between OPD and domain teachers. The provided LaTeX source contains tables labeled `tab:sci-enhanced-teacher`, `tab:long_sft_rl_results`, and `tab:main_results`. The text's reference to "Table 8" is ambiguous without the final PDF numbering, and the text fails to explicitly name the table containing the OPD results. If the final PDF numbering differs from the logical sequence implied by the text, this creates a disconnect for the reader trying to verify the claim. The text should reference the specific table content or ensure the numbering aligns perfectly with the final compilation.

Finally, while the baseline scores (47.4 for HLE) are consistent across Table 5 and Table 6, the text in Section 4.2.1 attributes the 47.4 -> 50.3 improvement specifically to the "search-enhanced teacher." A reader might confuse this with the "science-enhanced teacher" (Table 6) which also starts from 47.4. While not a fatal logical break, the text should explicitly state "the search-enhanced teacher improves the baseline HLE score from 47.4 to 50.3" to prevent ambiguity about which teacher achieved which result, ensuring the argument that "search training helps HLE" is clearly linked to the correct data point.

These issues are primarily textual inconsistencies (writing) rather than fundamental flaws in the scientific logic, as the tables themselves appear internally consistent. Correcting the numbers in the text and clarifying the table references will restore full logical consistency.

Action items.

- [**writing**] Section 4.2.1 claims GAIA score increases to 85.4, but Table 5 (search_rl.tex) shows 95.1. Correct the text to match the table value (95.1) or verify the table data.
- [**writing**] Section 4.2.1 states the search-enhanced teacher improves HLE from 47.4 to 50.3. Table 5 confirms this, but Table 6 (Science-enhanced) also lists 47.4 as baseline. Explicitly clarify in text that 47.4 is the shared baseline for all teachers to avoid confusion about which teacher achieved 50.3.
- [**writing**] Section 4.2.2 references 'Table 8' for OPD results, but the main results table is labeled `tab:main_results` (likely Table 4). Update text to reference the correct table label or ensure final PDF numbering matches the text's 'Table 8' claim.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes a bold rhetorical claim in its title and abstract: that a 35B model can "reach trillion-parameter performance" and "match" 1T models. While the results demonstrate impressive gains on specific long-horizon benchmarks (e.g., SEAL-0, HiPhO, FS-O), the evidence does not support a universal claim of parity. Table 1 explicitly shows the model trailing behind 1T baselines (GPT-5.5, Kimi-K2.6) on significant portions of the evaluation suite, particularly in engineering tasks (MLE-Bench-Lite: 43.9 vs 72.7) and general coding (SciCode: 44.3 vs 56.1).

The rhetoric of "reaching" or "matching" trillion-parameter performance suggests a general equivalence that the data does not support. The paper successfully argues that horizon scaling is a viable *alternative* for specific long-horizon, tool-use, and reasoning tasks, but it overgeneralizes this to imply a broader performance equivalence. The conclusion and abstract should be narrowed to reflect that the model matches 1T performance *on specific long-horizon benchmarks* rather than generally.

Furthermore, the paper frames the "failure" on engineering tasks as a specific limitation of the current training recipe ("MLE optimization is not a static problem-solving task") rather than a fundamental boundary of the horizon-scaling approach compared to parameter scaling. A more honest assessment would acknowledge that for tasks requiring deep internalized knowledge or complex, multi-stage engineering workflows, parameter scaling currently remains superior, and the "trillion-parameter" claim is conditional on the task type. The limitations section is present but too brief to adequately contextualize these significant performance gaps against the headline claims.

Action items.

- [**writing**] Title and Abstract claim 'Reaching Trillion-Parameter Performance' and 'match the performance of 1T models.' Table 1 shows the model is outperformed by 1T models (GPT-5.5, Kimi-K2.6) on 6 of 14 benchmarks (e.g., MLE-Bench-Lite: 43.9 vs 72.7; SciCode: 44.3 vs 56.1). Replace 'match' with 'compete with on specific long-horizon benchmarks' and qualify the title to reflect performance parity only on a subset of tasks, not a general equivalence.
- [**writing**] Abstract states the method 'reaches trillion-parameter-level performance' based on results from a single 35B model family (Qwen3.5 derivatives). The conclusion implies this scaling path is a general solution for 'any' agent. Add a limitation explicitly stating that the 'trillion-parameter' claim is specific to the tested benchmarks and model architecture, and that generalization to other domains or model families remains untested.
- [**writing**] The Introduction and Abstract frame the work as a 'practical path' to replace parameter scaling. However, the results show the method fails to close the gap on engineering tasks (MLE-Bench) where parameter scaling clearly wins. The narrative omits this failure mode. Add a sentence in the Limitations section acknowledging that for tasks requiring deep internal knowledge or complex engineering workflows, parameter scaling still outperforms horizon scaling in the current regime.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a 35B Mixture-of-Experts agent model trained on long-horizon trajectories across six domains (search, engineering, science, instruction following, tool calling, and general agentic tasks). From a safety and ethics perspective, the work does not exhibit foreseeable, non-trivial risks that are unaddressed.

The data pipeline described in Section 3 (data_pipeline.tex) relies on public benchmarks (e.g., MLE-Dojo, Kaggle competitions, WildChat-1M, NVIDIA Nemotron-RL) and synthetic data generation via self-play and tool-augmented search. The paper explicitly states that trajectories are collected in sandboxed environments or from public sources, and no Personally Identifiable Information (PII) or sensitive human-subjects data (e.g., medical records, private communications) is released or used without consent. The use of "simulated users" in tool-calling tasks (Section 3.5) is a standard synthetic data technique and does not constitute covert surveillance or deception of real individuals.

The model's capabilities (long-horizon planning, tool use, code generation) are dual-use by nature, as is common in the field of agentic AI. However, the paper does not describe a novel method that specifically lowers the barrier to a concrete harmful act (e.g., automated vulnerability exploitation, biological synthesis, or targeted disinformation generation) beyond the general capabilities of existing frontier models. The evaluation benchmarks (e.g., GAIA, HLE, SciCode) are standard academic tests for reasoning and tool use, not operational attack simulations. The authors do not release operational exploit code or actionable attack recipes; the "12-hour optimization run" (Section 5.3.1) is a benign machine learning engineering task.

There are no missing disclosures regarding human subjects, IRB approval, or data licensing violations. The paper acknowledges limitations in Section 6 but does not require a specific safety addendum for the risks identified, as the risks are inherent to the class of models being studied and are not exacerbated by specific, unmitigated methodological choices in this work. The verdict is `accept` with no action items required for this lens.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling narrative for scaling agent horizons, but the experimental design in Section 4 and Tables 1-3 contains critical gaps that prevent the evidence from fully supporting the headline claims of "trillion-parameter performance" and specific mechanism attribution.

First, the stability of the reported results is unverified. Table 1 presents single-point accuracy scores (e.g., 56.4 on SEAL-0, 80.6 on IFBench) without any measure of variance. For benchmarks with small test sets (IFBench has 294 prompts; MLE-Bench-Lite has 22 competitions), a single run is statistically insufficient to rule out sampling noise. A 1-2 point fluctuation is common in these settings, yet the paper presents these numbers as definitive improvements over baselines. The authors must report results averaged over at least 3-5 independent seeds with standard deviations or confidence intervals to demonstrate that the observed gains are robust and not artifacts of a lucky seed or a specific test set split.

Second, the attribution of performance gains to the proposed "Domain-Routed On-Policy Distillation" (OPD) is confounded. The comparison in Table 3 (Agents-A1 vs. Agents-A1-SFT) conflates the effects of the distillation algorithm, the salient vocabulary alignment (SVA) loss, and the domain routing mechanism. The paper claims the gain comes from "reducing conflicts caused by different reasoning patterns," but no ablation study isolates these components. For instance, does the gain persist if the student is distilled from teachers without the SVA loss? Or if the routing is removed? Without these controls, the 6-10 point improvements could be driven by the increased compute budget of the distillation phase or the specific SVA formulation rather than the routing mechanism itself.

Finally, the comparison against 1T-parameter models (Table 1) suffers from an unfair evaluation asymmetry. The authors evaluate their model with a custom, tool-augmented pipeline (search, visit, code, scholar tools) but rely on "official results" for the 1T baselines, which were likely generated under different, potentially less favorable, evaluation protocols. If the 1T models were evaluated with the same tool-augmented setup, the performance gap might narrow or reverse. To validly claim that a 35B model matches 1T performance, the comparison must be conducted under identical evaluation conditions, including the same tool usage, prompt engineering, and seed counts.

Action items.

- **[writing]** The paper presents a compelling narrative for scaling agent horizons, but the experimental design in Section 4 and Tables 1-3 contains critical gaps that prevent the evidence from fully supporting the headline claims of "trillion-parameter performance" and specific mechanism attribution. First, the stability of the reported results is unverified. Table 1 presents single-point accuracy scores (e.g., 56.4 on SEAL-0, 80.6 on IFBench) without any measure of variance. For benchmarks with small test s

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in this paper is generally consistent with common practices in large-scale LLM benchmarking (reporting point estimates), but it lacks the necessary uncertainty quantification to support the strong comparative claims made in the text.

Specifically, Section 4.1 mentions that MLE-Bench-Lite results are averaged over three seeds, yet Table 1 and Table 3 present only single point estimates (e.g., 43.9) without standard deviations, standard errors, or confidence intervals. In the absence of variance metrics, it is impossible to determine if the reported improvements over baselines (e.g., 43.9 vs 34.9) are statistically significant or within the noise of the evaluation. The same issue applies to other benchmarks where multiple seeds are implied or used but not reported.

Furthermore, the paper makes numerous claims of "outperforming" or being "significantly better" than baselines across a large matrix of benchmarks (16 benchmarks $\times$ 6 baselines = 96 comparisons). No multiple-comparison correction (such as Bonferroni or Benjamini-Hochberg) is applied to control the false discovery rate. With this many tests, the probability of observing at least one "significant" difference by chance alone is high. The authors should either apply a correction to the reported p-values (if tests were run) or explicitly acknowledge the lack of correction and frame the results as exploratory.

Finally, Figure 2 includes a visual representation of variance ("shaded band"), but the text and tables do not provide the numerical values or the specific statistical measure (e.g., SD, 95% CI) used to generate this band. This disconnect between the visual and the quantitative reporting reduces the interpretability of the variance claim.

These are primarily reporting issues (writing) that can be fixed by adding standard deviations to tables and clarifying the statistical basis for "outperforming" claims, though the lack of multiple-comparison correction touches on the validity of the comparative conclusions (science).

Action items.

- [**writing**] The statistical reporting in this paper is generally consistent with common practices in large-scale LLM benchmarking (reporting point estimates), but it lacks the necessary uncertainty quantification to support the strong comparative claims made in the text. Specifically, Section 4.1 mentions that MLE-Bench-Lite results are averaged over three seeds, yet Table 1 and Table 3 present only single point estimates (e.g., 43.9) without standard deviations, standard errors, or confidence intervals. In

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and readable, with a clear narrative flow from the problem statement to the proposed solution and results. However, there are several instances of grammatical errors, sentence fragments, and weak transitions that disrupt the reader's momentum, particularly in the Experimental Results section.

In Section 3.1, the explanation for the performance drop in full-domain SFT is presented as two disjointed sentences, the second of which lacks a clear connection to the first. This forces the reader to infer the causal link. Similarly, in Section 3.2, the phrase "due to that" is a grammatical error that should be corrected to "because" or "due to the fact that." The use of "Besides" as a transition to introduce the case study analysis is weak and slightly informal for a technical paper; a stronger transition like "Furthermore" or "To further investigate" would improve the flow.

Additionally, there are minor issues with sentence fragments and ambiguous phrasing. For instance, the sentence "On Vita, MatTools, and MolBench-MS Binding Affinity tasks..." is slightly awkward and could be clearer. The transition between the description of the 12-hour optimization run and the analysis of the figure in Section 4.1 is also somewhat abrupt.

These issues are fixable through careful editing and do not reflect on the scientific merit of the work. Addressing them will significantly improve the clarity and professionalism of the manuscript.

Action items.

- [**writing**] The paper is generally well-structured and readable, with a clear narrative flow from the problem statement to the proposed solution and results. However, there are several instances of grammatical errors, sentence fragments, and weak transitions that disrupt the reader's momentum, particularly in the Experimental Results section. In Section 3.1, the explanation for the performance drop in full-domain SFT is presented as two disjointed sentences, the second of which lacks a clear connection to t